

# Outline

## 1. Potential outcome framework

causal effect

## 2. Experiment { CRE, SRE, MPE Fisherian / Neymanian Design / Analysis

## 3. Observational { Identification of ATE, ATT Estimation: outcome / IPW / DR Matched Observational study

## 4. Sensitivity Analysis { E-value ACE Rosenbaum's

## 5. IV.

## 1. (1) Science table

| $i$      | $Y_i(1)$ | $Y_i(0)$ |
|----------|----------|----------|
| 1        | $Y_1(1)$ | $Y_1(0)$ |
| 2        | $Y_2(1)$ | $Y_2(0)$ |
| $\vdots$ | $\vdots$ | $\vdots$ |
| $n$      | $Y_n(1)$ | $Y_n(0)$ |

## (2) causal effects

### (i) individual treatment effect

$$\tau_i = Y_i(1) - Y_i(0)$$

### (ii) ALE / ATE

$$\begin{aligned}\tau &= \frac{1}{n} \sum_{i=1}^n \tau_i \\ &= \frac{1}{n} \sum_{i=1}^n (Y_i(1) - Y_i(0)) \\ &= \frac{1}{n} \sum_{i=1}^n Y_i(1) - \frac{1}{n} \sum_{i=1}^n Y_i(0)\end{aligned}$$

### (iii) Other estimates: OR, RR

binary outcome :  $Y_i(1), Y_i(0) \in \{0, 1\}$

$$\Rightarrow RR = \frac{\Pr(Y_i(1)=1)}{\Pr(Y_i(0)=1)} = \frac{p_1}{p_0}$$

$$OR = \frac{p_1 / (1 - p_1)}{p_0 / (1 - p_0)}$$

(3) observed data

$$Y_i = \begin{cases} Y_i(1), & \text{treated} \\ Y_i(0), & \text{o.w.} \end{cases}$$
$$= Z_i Y_i(1) + (1 - Z_i) Y_i(0)$$

Chapter 3.

(1) CRE (treatment mechanism)

$n$        $n_1$

$$\Pr(Z=z) = \frac{1}{\binom{n}{n_1}} \quad \text{permutational}$$

(2) FRT

(i) Null hypothesis

Strong/Sharp null

$$H_{0F}: Y_i(1) = Y_i(0) \quad \forall i$$

(ii) randomization distribution

$T(Z, Y)$  uniform distribution over  $R \subset \binom{n}{n_1}$

$T(Z^r, Y)$ ,  $Z^r$  is taken over all permutations of observed  $Z$

(iii) procedure of FRT:

permutation test

choice of test statistics:

$t$ , studentized  $t$ , wilcoxon, KS

(iv.) finite sample exactness

$$\Pr(\hat{p}_{FRT} \leq \alpha) \leq \alpha$$

$$E(\hat{p}_{FRT}) = p_{FRT}$$

Ch4, Neymanian inference in CRE

(1) Null hypothesis: weak null

$$H_0: \bar{Y}(1) = \bar{Y}(0)$$

(2) point estimator:  $\hat{\tau} = \hat{\bar{Y}}(1) - \hat{\bar{Y}}(0)$

$$\text{variance estimator: } \hat{V} = \frac{\hat{S}(1)^2}{n_1} + \frac{\hat{S}(0)^2}{n_0}$$

$$E(\hat{\tau}) = \tau$$

$$E(\hat{V}) = \frac{S(1)^2}{n_1} + \frac{S(0)^2}{n_0}$$

$$\geq \frac{S(1)^2}{n_1} + \frac{S(0)^2}{n_0} - \underbrace{\left[ \frac{S^2(\tau)}{n} \right]}_{\rightarrow \text{can't estimate}}$$

$$= \text{var}(\hat{\tau})$$

$$\hat{\bar{Y}}(1) = \frac{1}{n_1} \sum_{i=1}^{n_1} Z_i Y_i$$

(3) Asymptotic distribution

$$\frac{\hat{\tau} - \tau}{\sqrt{\hat{V}}} \xrightarrow{d} N(0, 1)$$

$$\hat{V} - E(\hat{V}) \xrightarrow{p} 0$$



(4) regression analysis

$$\ln(Y \sim 1 + z) \\ \nwarrow \hat{z} \\ \hat{V}_{\text{FW}}$$

Ch5 SRE K strata

(1) treatment mechanism

$$n_{[k]}, n_{[k]1}$$

$$Pr(z = z) = \frac{1}{\binom{n_{[k]1}}{n_{[k]}}}$$

$$(2) \text{ AUE: } z = \sum_{k=1}^K \pi_{[k]} \bar{z}_{[k]}, \quad \pi_{[k]} = \frac{n_{[k]}}{n}$$

(3) FRT: permute  $z$  within stratum

$$T(z, Y)$$

(4) Neymanian

$$\hat{z} = \sum_{k=1}^K \pi_{[k]} \hat{z}_{[k]}$$

$$\hat{V} = \sum_{k=1}^K \pi_{[k]}^2 \hat{V}_{[k]}$$

$$\frac{\hat{z} - z}{\sqrt{V}} \xrightarrow{d} N(0, 1)$$

## (5) post-stratification in CRE

|          | Design          | Analysis              |
|----------|-----------------|-----------------------|
| discrete | SRE             | post-stratification   |
| X        |                 |                       |
| general  | Rerandomization | regression adjustment |

## Ch 6. (1) Rerandomization

(i) Why? covariate balance

(ii) procedure? (ReM)

$$M = \hat{z}_x^T \text{cov}(X)^{-1} \hat{z}_x \in \mathbb{R}$$

①  $z \sim \binom{n}{n_1}$

② compute  $\hat{z}_x, M$

③ Accept iff  $M < a$

④ otherwise reject  $z$   
and go to ①

(iii) inference

(2) Regression adj.

(i) Fisher's ANCOVA

$$\text{lm}(Y \sim 1 + Z + X)$$

↑  
 $\hat{z}$

(ii) Lin's estimator

center  $X$

$$\ln(Y_i | 1 + \underbrace{Z_i}_{\hat{Z}_L} + X_i + \underbrace{Z_i X_i}_{\hat{V}_{FW}})$$

$$\textcircled{1} Y_{i(1)} - X_i^\top \hat{\beta}_1$$

$$Y_{i(0)} - X_i^\top \hat{\beta}_0$$

② projection / prediction

(iii). extension to SRE

$$\hat{Z}_L = \sum_{k=1}^K \pi_{[k]} \hat{Z}_{[k], L}$$

$$\hat{V}_L = \sum_{k=1}^K \pi_{[k]}^2 \hat{V}_{[k], L}$$

## Ch7 MPE

(1) treatment mechanism SRE

$$\textcircled{1} \text{SRE} \Rightarrow n_{[k]} = 2, n_{[k]1} = n_{[k]0} = 1$$

$$\textcircled{2} Z_i = \begin{cases} 1, & \text{unit } i \text{ to be treatment} \\ 0, & \text{o.w.} \end{cases}$$

$$\textcircled{3} Y_{i1}, Y_{i0}$$

$$\begin{aligned} \hat{Z}_i &= \begin{cases} Y_{i1} - Y_{i2}, & \text{if } Z_i = 1 \\ Y_{i2} - Y_{i1}, & \text{if } Z_i = 2 \end{cases} \\ &= (2Z_i - 1)(Y_{i1} - Y_{i2}) \end{aligned}$$

(2) FRT (i) procedure: randomize the signs

choices:  $t$ , wilcoxon, signstat, MacNeymar

- (ii) covariate adjustment  
 < ① use residuals  
 ② model fitting

(3) Neymanian

$$(i) \bar{z} = \frac{1}{n} \sum_{i=1}^n z_i$$

$$\hat{V} = \frac{1}{n-1} \sum_{i=1}^n (z_i - \bar{z})^2$$

(ii) regression implementation:

$$\text{lm}(\hat{z}_i \sim 1) \quad \hat{z}_i \quad \hat{V}_{OLS}$$

- (iii) covariate adj.  
 $\hat{z}_{adj} = \bar{z} - \hat{\gamma}^T \hat{x}$   
 $\text{lm}(\hat{z}_i \sim 1 + \hat{x}_{i,n})$   
 $\hat{z}_{adj}$

Part II (read Ch8, Ch9 first)

Ch10 observational study

(1) observational data

$$(Y_i(1), Y_i(0), Z_i, X_i) \stackrel{iid}{\sim} (Y(1), Y(0), Z, X)$$

(2) nonparametric identification

$$\bar{z} = E\{Y(1) - Y(0)\}$$

ignorability  $Z \perp\!\!\!\perp \{Y(1), Y(0)\} \mid X$

$$E(Y(1)) = E(E(Y(1) \mid X)) = E(E(Y(1) \mid X, Z=1)) = E(E(Y \mid X, Z=1)) = E(E(Y \mid X, Z=1))$$

## Ch 11

(1)  $e(x) = \Pr(Z=1|X) = E(Z|X)$

(2) why important?

(i) dimension reduction tool

$$Z \perp\!\!\!\perp \{Y(1), Y(0)\} | X \Rightarrow Z \perp\!\!\!\perp \{Y(1), Y(0)\} | e(X)$$

(ii) Propensity score weighting

$$E(Y(1)) = E\left(\frac{ZY}{e(X)}\right), E(Y(0)) = \dots$$

(iii) balancing score

$$Z \perp\!\!\!\perp X | e(X)$$

## Ch 12 Doubly robust / AIPW estimator

$$(1) \hat{\mu}_1^{dr} = \frac{1}{n} \sum_{i=1}^n \left[ m_1(X_i, \hat{\beta}_1) + \frac{Z_i \{Y_i - m_1(X_i, \hat{\beta}_1)\}}{e(X_i, \alpha)} \right]$$

$$\hat{\mu}_1^{dr} = E[Y(1)] \text{ if either } m_1(X, \beta) = m_1(X) \\ \text{or } e(X, \alpha) = e(X)$$

## Ch 13 ATT

$$(1) E(Y(1) - Y(0) | Z=1) = E(Y|Z=1) - E(Y(0)|Z=1)$$

(2) nonpara id

$$E(Y(0)|Z=1) = E(E(Y(0)|X, Z=1) | Z=1) = E(E(Y|X, Z=0) | Z=1)$$

$\hookrightarrow E(Y(0)|X, Z=0) = E(Y|X, Z=0)$

### (3) IPW regression DR

$\hat{M}_0^{\text{or}}$

Ch 14

$$\tau_h = \frac{E(h(x)z(x))}{E(h(x))}$$

pscore in regression

(1)  $e(x)$  as a covariate

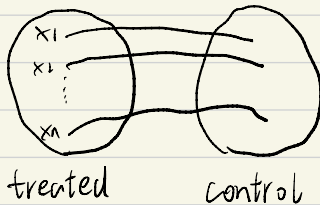
$$\ln(Y_i | 1 + \underbrace{z_i}_{\hat{z}_e} + e(x_i))$$

$$\tau_e = \tau_0 = \frac{E\{h_0(x)z(x)\}}{E(h_0(x))}$$

$$h_0(x) = e(x)\{1 - e(x)\}$$

(2) regression weighted by inverse of  $e(x)$

### Ch 15 Matched observational study



(1) setup: Match  $x_i$  in treated group  
with  $m(i)$  units in the control group

$$(2) \hat{\tau}^m = \frac{1}{n} \sum_{i=1}^n (\hat{Y}_i(1) - \hat{Y}_i(0)) \rightarrow \text{is biased}$$

$$\text{if } z_i = 1, \quad \hat{Y}_i(1) = Y_i$$

$$\hat{Y}_i(0) = \frac{1}{n} \sum_{\substack{j \in J(x_i) \\ z_j = 0}} Y_j$$

$$(M) \quad \hat{\tau}^{mbc} = \hat{\tau}^m - \hat{B}$$

$$= \frac{1}{n} \sum_{i=1}^n \hat{\psi}_i$$

$$\hat{\psi}_i = \hat{M}_1(x_i) - \hat{M}_0(x_i) + 2(z_i - 1) \left(1 + \frac{k_i^2}{n}\right) (Y_i - \hat{M}_{2i}(x_i))$$

$$(3) \hat{\tau}^{reg} = \frac{1}{n} \sum_{i=1}^n (\hat{M}_1(x_i) - \hat{M}_0(x_i))$$

$$\hat{\tau}^{dr} = \hat{\tau}^{reg} + \frac{1}{n} \sum_{i=1}^n \left( \frac{z_i \hat{R}_i}{\hat{E}(x_i)} - \dots \right)$$

$$\hat{\tau}^{mbc} = \hat{\tau}^{reg} + \frac{1}{n} \sum_{i=1}^n \left[ z_i \left(1 + \frac{k_i}{n} \hat{R}_i - \square \right) \right]$$

(4) ATT